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Theoretical characterization of Al(III) binding to KSPVPKSPVEEKG: Insights into the propensity of aluminum to interact with key sequences for neurofilament formation

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ABSTRACT

Classical molecular dynamic simulations and density functional theory are used to unveil the interaction of aluminum with various phosphorylated derivatives of the fragment KSPVPKSPVEEKG (NF13), a major multi-phosphorylation domain of human neurofilament medium (NFM). Our calculations reveal the rich coordination chemistry of the resultant structures with a clear tendency of aluminum to form multidentate structures, acting as a bridging agent between different sidechains and altering the local secondary structure around the binding site. Our evaluation of binding energies allows us to determine that phosphorylation has an increase in the affinity of these peptides towards aluminum, although the interaction is not as strong as well-known chelators of aluminum in biological systems. Finally, the presence of hydroxides in the first solvation layer has a clear damping effect on the binding affinities. Our results help in elucidating the potential structures than can be formed between this exogenous neurotoxic metal and key sequences for the formation of neurofilament tangles, which are behind of some of the most important degenerative diseases.

1. Introduction

In spite of its abundance in the Earth's crust, [1,2] aluminum is excluded from biotic cycles. However, due to its versatility in various technological applications, the bioavailability of aluminum has markedly increased in the last century. This fact is unlikely to be without consequences, with increasing evidence of the toxic effects of aluminum in biological systems, [3–5]. In fact, the presence of aluminum in the human body has been linked with several diseases, [6,7] such as dialysis encephalopathy, osteodystrophy and microcytic anemia derived from chronic renal failure wherein the aluminum accumulates in the kidney, liver, bone, and heart. [8,9] The full molecular bases of aluminum toxicity are still unknown, but some potentially mechanisms have been outlined. For instance, it has been well established that aluminum shows a significant pro-oxidant activity in biological systems, [10–12] it also inhibits the normal function of various enzymes involved in the glycolysis pathway [13–16] and in the production of glutamate [17,18] affecting the TCA cycle. [19–21] Moreover, several studies have shown that aluminum competes with magnesium as a metal-ATP co-factor [22–27] and that it can modify the electronic and structural properties of the chelated biological ligands. [28–30] One has to take into account that aluminum has a predominance for O donor species, [31] with negatively charged groups, such as carboxylates, phenolates, catecholates, and phosphates. [32–34]

Aluminum is also considered as a neurotoxic element associated with Alzheimer's disease (AD), but it is not clear the role that aluminum plays in neurodegeneration. [35–39] In a seminal study, Exley et al. [40] demonstrated the ability of aluminum to bind β-amyloids, a result confirmed later computationally. [41] Moreover, in vitro experiments determined that Al(III) promotes the aggregation of β-amyloid peptides more efficiently than other metals, [42–44] with the main constituent of insoluble amyloid plaques known as senile plaques, and in fact, aluminum has been detected in senile plaques extracted from the brains of patients with AD. [45]

Besides, this metal also induces the abnormal neurofilament tangle (NFT) aggregation and promotes the hyperphosphorylation of normal proteins. [46,47] As a matter of fact, phosphorylation of peptides is thought to increase their affinity for aluminum and could be a key

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aspect in understanding the aluminum promotion on the NFT aggregation process. [48,49]

The NFT's are composed of several proteins as the τ or the NF; the latter represents the main cytoskeleton elements in mature neurons and according to their molecular weights they are classified as low, medium and high. The determination of the amino acid sequence of the human neurofilament medium (NFM) showed that its C-terminal domain has the fragment KSPVPKSPVEEKG (NF13), which is repeated six times serially and is the major multiphosphorylation domain of the NFM, with the KSPV tetramer occurring 12 times. [50–53]

The study of coordination complexes with multivalent metals is very challenging. Within the development of force fields, a great effort has been made to incorporate more information into the different potentials used to simulate biometal compounds, [54,82–86] considering important aspects related to the nature of the metal ion, such as the coordination numbers, the oxidation states, and the chemical bonding, among others. In this work, we use a theoretical protocol that combines quantum mechanics calculation, via density functional theory, to consider explicitly the electronic structure of the aluminum-peptide complexes, and classical molecular dynamics to explore their conformational space. As detailed below, we investigate the Al(III) coordination to the NF13 peptide, the role of phosphorylation in the increase of the binding affinity to aluminum, and the structural changes caused by the presence of the metal. To do so, three phosphorylated variants of the NF13 peptide were considered: two of them with one serine phosphorylated (S2P or S7P) and the other with both serines phosphorylated (S(2,7)P). Different coordination modes were tested, where the aluminum is coordinated to the peptide in a mono, bi, and tridentate fashion, with various combinations between the phosphoserines and glutamate residues. Table 1 lists the binding sites in the phosphorylated peptides. Finally, we report estimations of the aluminum binding energies for the different coordination modes. Our results indicate a significant increase in the aluminum affinity for these peptides upon phosphorylation, and highlight the ability of aluminum to act as an aggregation factor between different sidechains in the peptide, altering its local secondary structure. The role of pH is also discussed by introducing different number of hydroxides in the first solvation layer, predicting a lower capacity of these peptides to bind aluminum as pH is raised.

2. Methodology

The structures were built from scratch with Molden [55] in a β-conformation, with an acetylated and amidated N and C-terminal, respectively. As in previous works, [56,57] molecular dynamics (MD) simulations were carried out using Gromacs package, [58,59] version 4.5.7 and the CHARMM36 force field, [60] including the non-bonded parameters of Al(III) from Faro et al., [61] who parameterized a simple Lennard-Jones plus Coulomb potential to reproduce the first and second hydration shells, as well as the enthalpy of hydration, the reorientational dynamics, and the vibrational spectral features with good agreement with experimental data. The peptides were solvated in a 3.75 nm^3 dodecahedral water box with the TIP3P model, [62] and depending on the case sodium and chloride ions were added to neutralize the charge. We followed the same protocol for preparing and running the simulations for all molecular systems. An initial energy minimization of the systems was performed using the steepest descent method for 50,000 steps. The Particle Mesh Ewald method [63,64] was used to calculate the electrostatic interactions with a cut-off of 0.8 nm for the short range van der Waals interactions. The integration time step was of 2 fs and after 1 ns equilibration, trajectories of 30 ns were performed in the canonical ensemble at 300 K with the Bussi thermostat. [65] All trajectories were sampled every 10 ps and the structures were clustered by the Gromacs utility that uses the algorithm described by Daura et al. [66] In our case, this clustering brings together more than 49% of the sampled structures into the first five groups.

The binding affinities were computed with the TURBOMOLE program; [67] one structure from the most populated cluster of each coordination mode was optimized using the B3LYP [68,69] functional and the 6-31G* basis set, the zero-point vibrational energy (ZPVE) and the thermal vibrational corrections at 298 K to evaluate the enthalpies and Gibbs free energies, were extracted from the frequency analysis; the entropy was computed using the standard statistical mechanics expressions for the partition function in the canonical ensemble. Solvent effects were included using the implicit solvation model COSMO. [70]

Table 1
Residues involved in the different coordination modes of the Al(III) to three variants of the KSPVPKSPVEEKG peptide; two with just one serine phosphorylated (S2P or S7P) and other with both serines phosphorylated (S(2,7)P). *X* is 2 or 7 depending on the case.

Coordination	Al@		
Mode	S2P or S7P	S(2,7)P	
Monodentate	S *X*	S2	S7
Bidentate		S2S7	
	S *X*E10	S2E10	S7E10
	S *X*E11	S2E11	S7E11
	E10E11	E10E11	
Tridentate		S2S7E10	S2S7E11
	S*X*E10E11	S2E10E11	S7E10E11

3. Results

The optimized structures of the selected free peptides and their aluminum complexes are displayed in Figs. 1, 2 and 3.

3.1. Conformational analysis from MD simulations

To analyze the effects of the coordination of aluminum in the conformation of the peptides, we calculated the Ramachandran angles from the MD trajectories of the free peptides and the aluminum complexes (see scatter plots in Figs. 4 and 5). While the starting structures for the simulations were generated using a β-sheet conformation, we find that either during the preparation or production phases some peptides explore the β-sheet ($-135°$, $135°$), PPII ($-75°$, $150°$) and both left- ($70°$, $25°$) and right-handed ($-70°$, $-30°$) α-helical regions. Within the relatively short timescales of our simulations, aluminum ligand exchanges were not observed along the simulation time, and we cannot expect that we obtain a full exhaustive sampling of the conformational space, but rather a local exploration in the vicinity of the initial states. That is precisely what we find when we look at the distributions of Ramachandran angles at the residue level (see SI Figs. 1–4), where we see density around the initial configuration for the central amino acid residues in the polypeptide chain and broader sampling for the terminal residues. We also identify trends for greater structuring of the peptide onto α-helix or α_L states with higher coordination to the Al. This is expected, as the more contorted helical dihedral angles will be required for the short polypeptide chain to coordinate to the Al cation. However, considerably longer simulation time-scales will be required to make these correlations quantitative, and hence we do not pursue this goal here.

Within the limits of the local sampling of our MD simulations (see below), we can check for changes in the rigidity of the peptides due to the presence of aluminum cation. In Fig. 6, the plots of the root mean square fluctuations (RMSF) of Cα's as a function of the residue in each of the phosphorylated complexes are depicted.

In the case of the NF13_S2P complexes (Fig. 6**A**), the coordination of the aluminum to the phosphoserine and the glutamic acids (S2P_5) shows a lowering of the RMSF's with respect to the free peptide, while in the monodentate complex the backbone remains almost unchanged (S2P_1), unlike of the coordination only with the glutamic acids

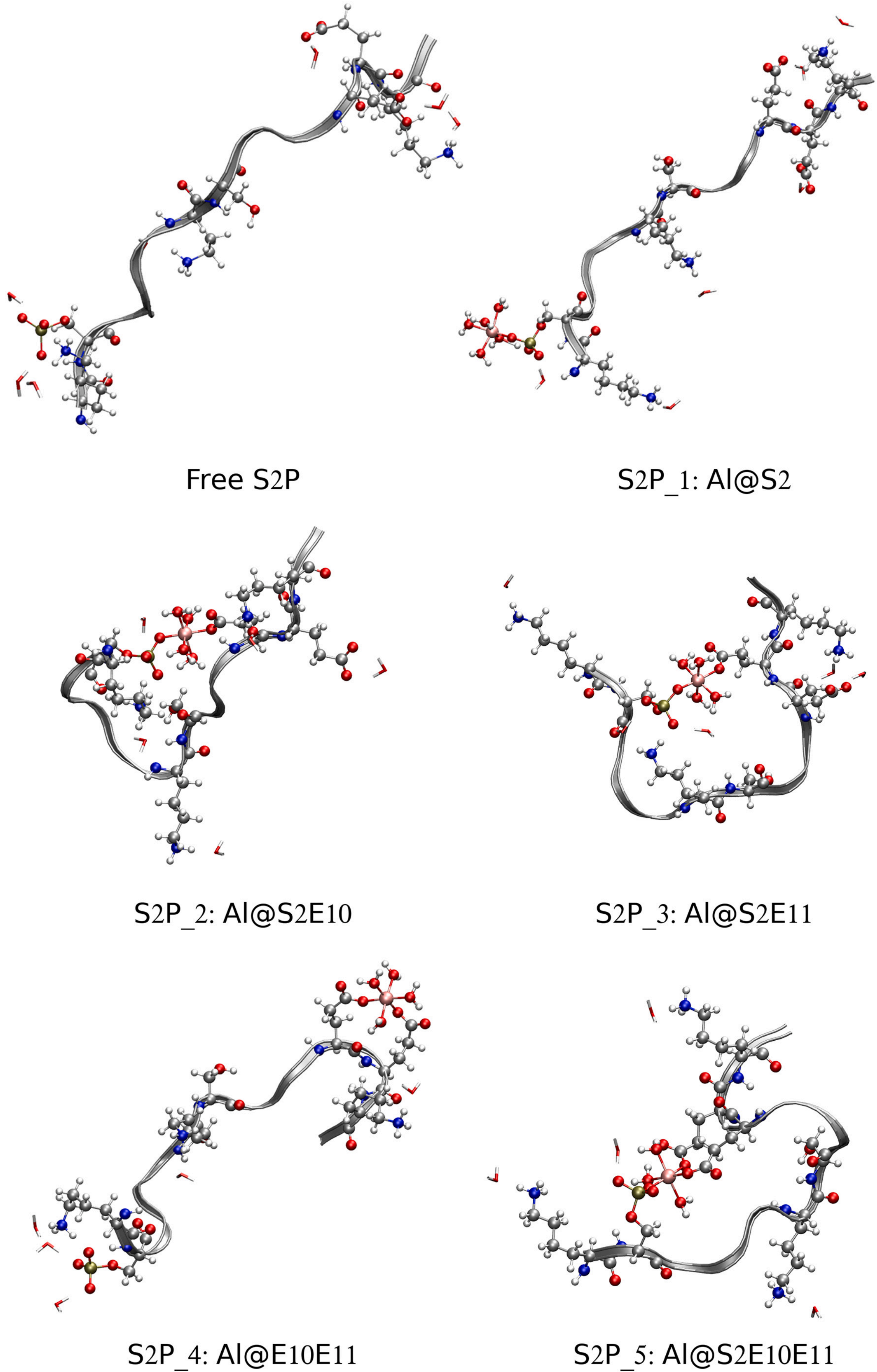

Fig. 1. Optimized structures of the NF13_S2P aluminum complexes.

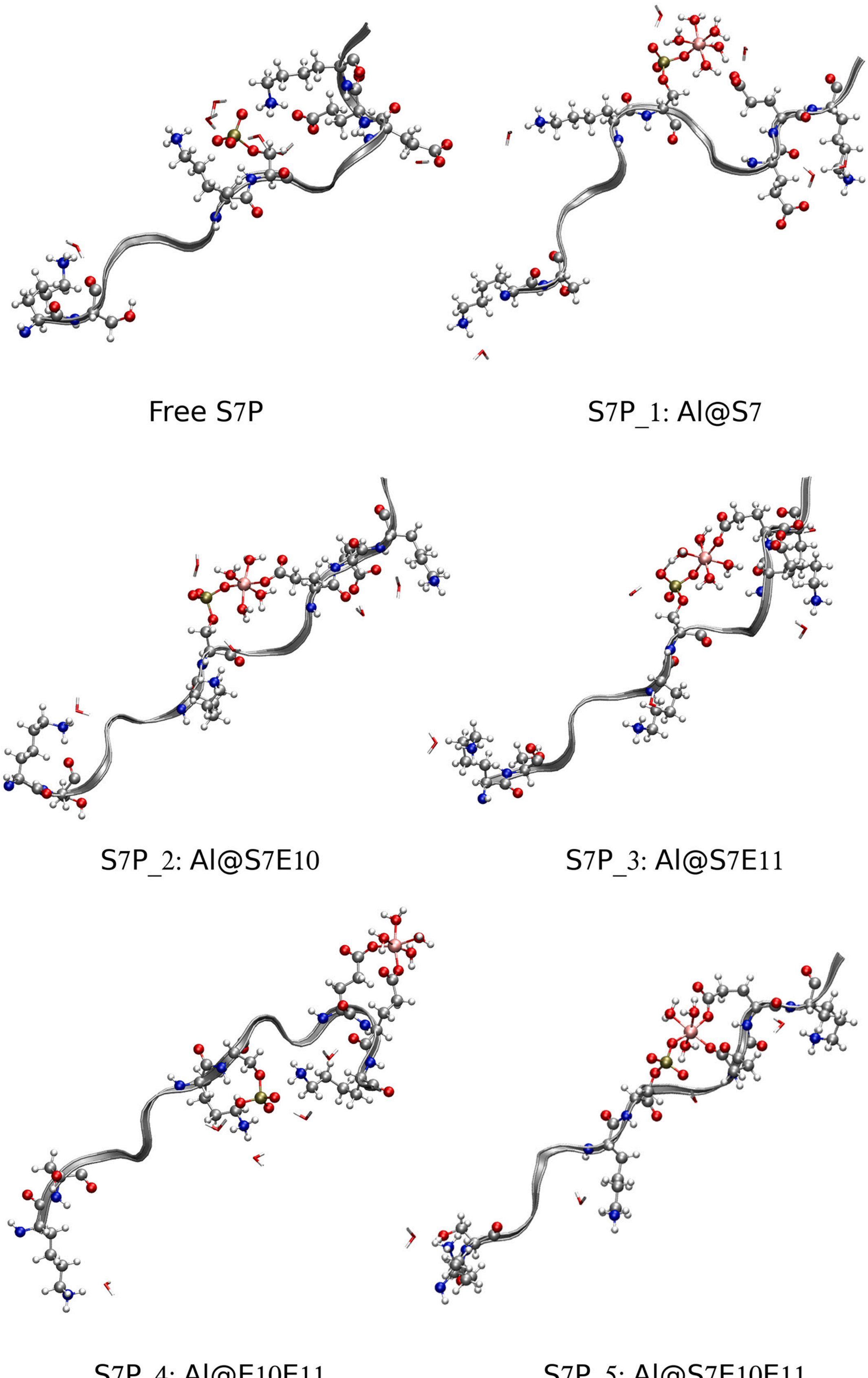

Fig. 2. Optimized structures of the NF13_S7P aluminum complexes.

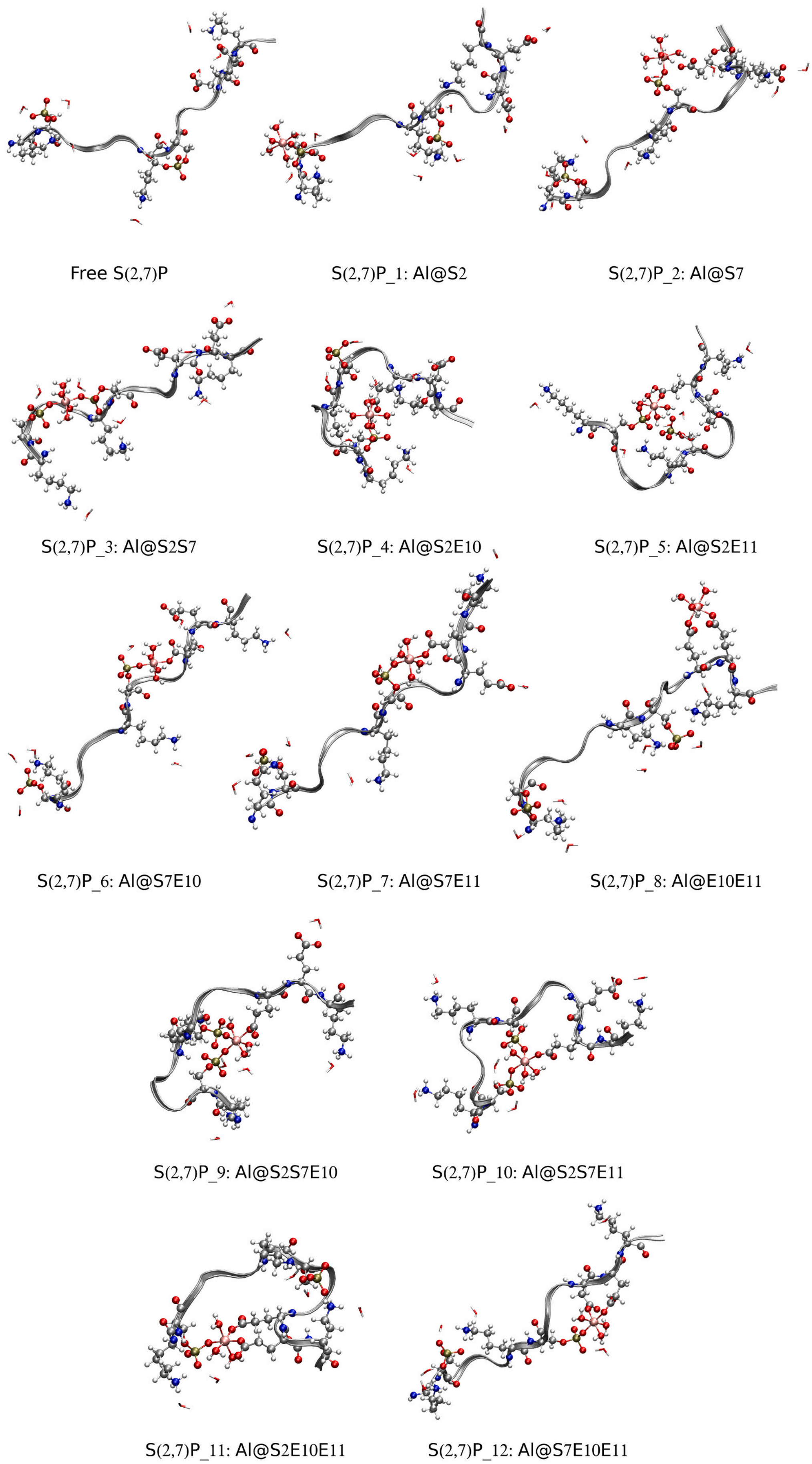

Fig. 3. Optimized structures of the NF13_S(2,7)P aluminum complexes.

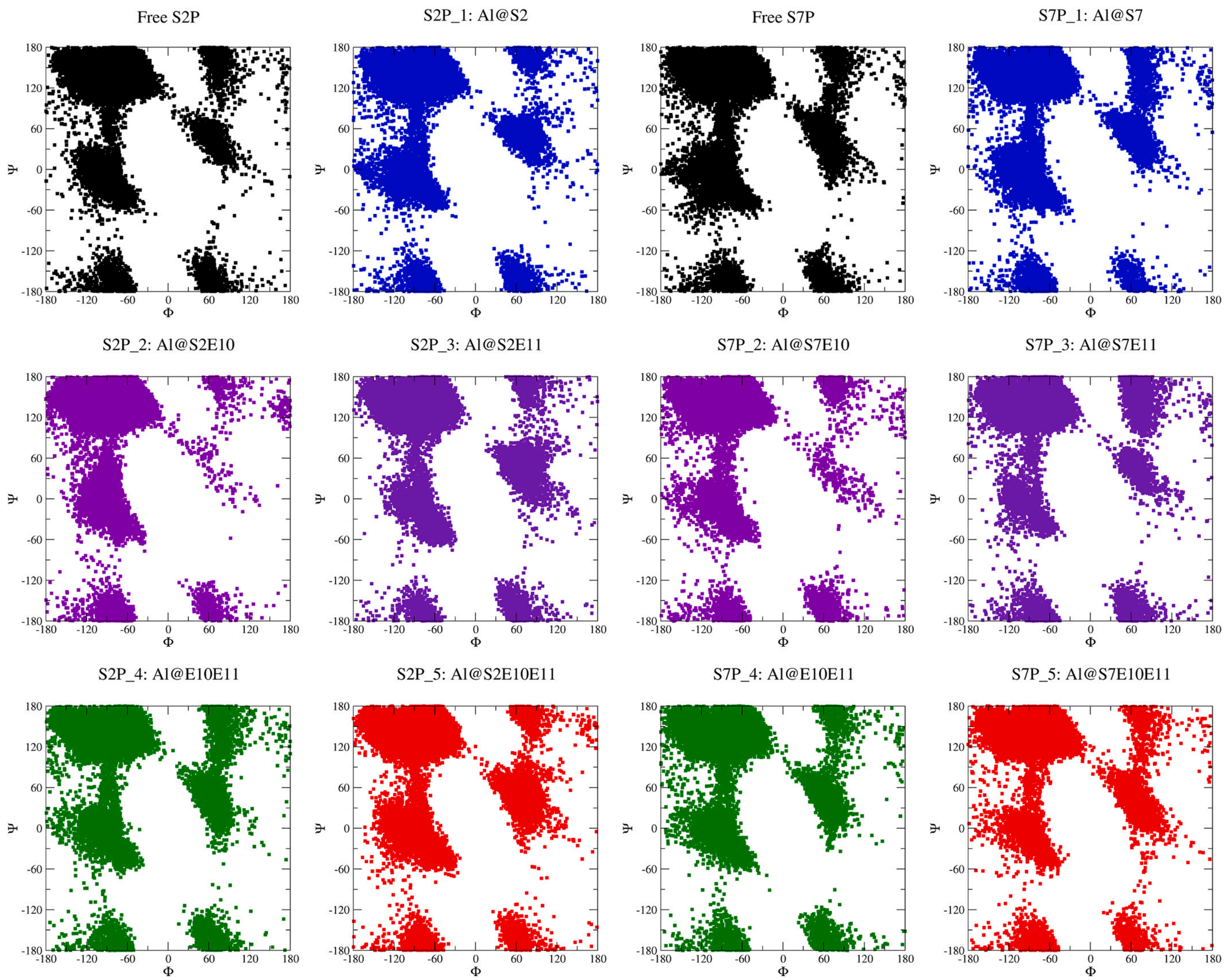

Fig. 4. Ramachandran plots from the MD simulations with the NF13_S2P (two left panels) and NF13_S7P (two right panels) systems. Free peptides (black), aluminum complexes monodentates (blue), bidentates (violet, indigo and green), and tridentates (red).

(S2P_4), that is more flexible. All these changes in the rigidity of the peptide due to the aluminum complexation are evident with the selected optimized structures (Fig. 1), were we observed that the rigidity of the backbone in the monodentate complex is quite similar to the free peptide; while there is a more rigid situation in the multidentate modes with the phosphoserine, in which a specific folding is required to coordinate the aluminum.

In the NF13_S7P complexes the changes are less clear (Fig. 6B); in the monodentate complex the flexibility is slightly greater than in the free peptide (S7P_1), probably due to the interactions of the side chains of the neighboring residues with the coordination site located in the middle of the peptide, as can be seen in the optimized structure (Fig. 2 S7P_1), where the lysine side chains move away from the phosphoserine respect to the free peptide. When aluminum binds the phosphoserine and one or two glutamic acids (S7P_2, S7P_3 and S7P_5), there is a decrease in the flexibility of the second half of the peptide, mainly. It is worth to note that the binding to E11 leads to a more rigid situation that when binding to E10, as in the previous case. The optimized structures of these phosphoserine multidentate modes (Fig. 2), show that the backbone is not altered dramatically due to the closeness of the binding sites, although it restricts the mobility of that part of the peptide. On the other hand, the coordination to both glutamic acids only alters the vicinity of the coordination site (S7P_4).

The results for the complexes in which both serines are phosphorylated, for clarity, can be found in Fig. 6**C** for mono and tridentate coordination modes, and in Fig. 6D for bidentate ones. In the case of the monodentate modes (S(2,7)P_1 and S(2,7)P_2), the peptide is just as rigid in the vicinity of the coordination site as in the free peptide, being the acid region slightly more flexible. However, most of the tridentate modes are more rigid than the free peptide, except in the case of the S(2,7)P_12 complex. These effects can be observed in the optimized structures (Fig. 3), where the peptides are folded into a loop when the binding residues are far away from each other. For the bidentate cases, the coordination to both phosphoserines increase the rigidity of that region without altering the acid part (S(2,7)P_3), due to the fact that these sidechains do not interact with the coordination site (Fig. 6D). The combination of S2P with each of the glutamic acids (S(2,7)P_4 and S(2,7)P_5) decrease the flexibility of the peptide, due to the formed loop (Fig. 3), unlike the respective combinations with S7P (S(2,7)P_6 and S(2,7)P_7), where the flexibility is slightly higher in the first half of the peptide and the changes are in the acid region. As expected, the coordination to both glutamic acids is the most flexible and the drastic changes in the acid region are because the side chain of the K12 tends to interact with the S7P (Fig. 3).

In summary, we can say that the complexation of aluminum to the phosphorylated peptides changes their conformations depending on the

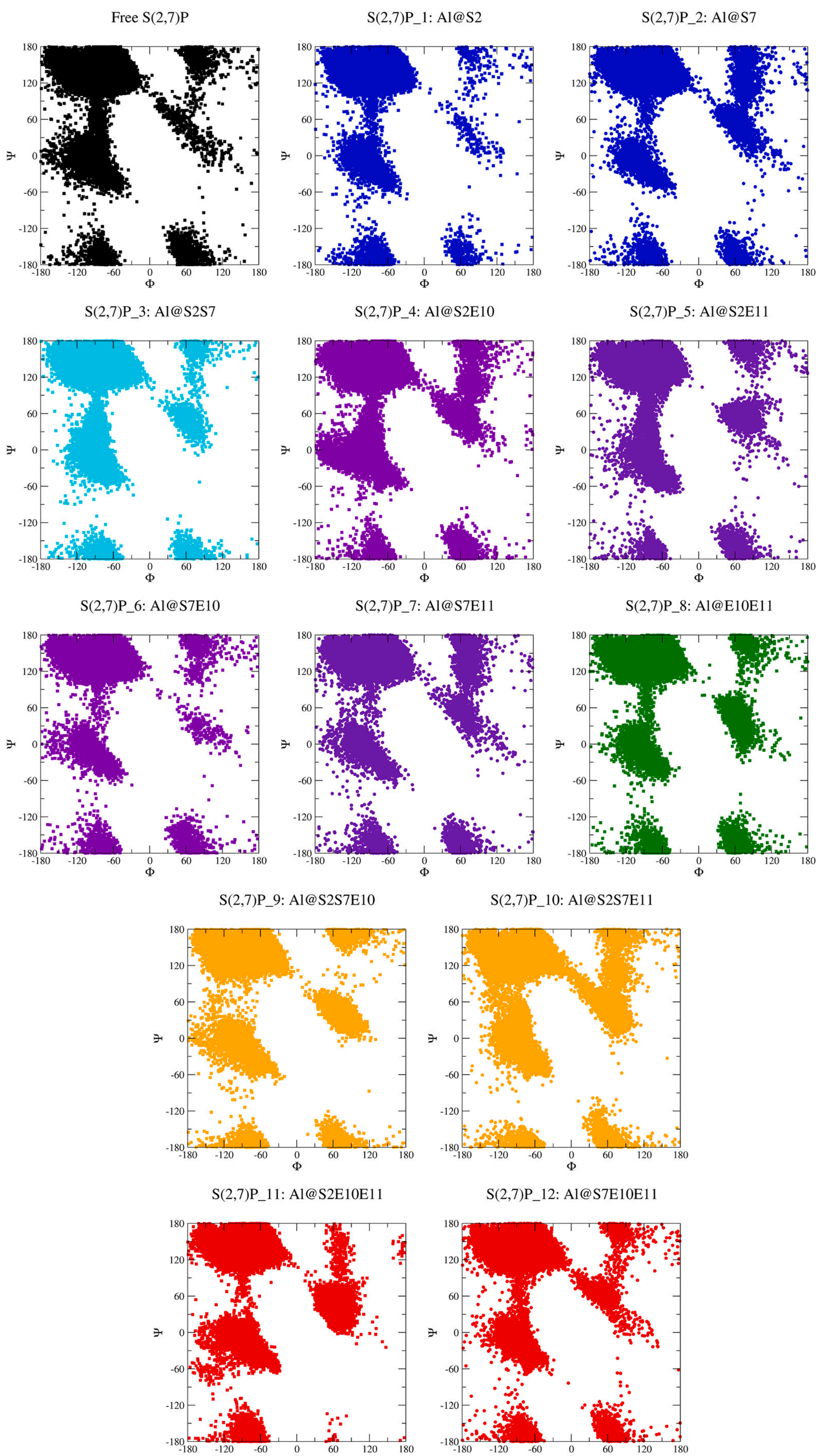

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Fig. 5. Ramachandran plots from the MD simulations with the NF13_S(2,7)P systems. Free peptide (black), aluminum complexes monodentates (blue), bidentates (cyan, violet, indigo and green), and tridentates (orange and red).

coordination mode. In general, the tridentate complexes tend to be more rigid, as opposed to coordination to both glutamic acids which is the most flexible. Besides, in the monodentate complexes, the coordination to S7P is affected by the proximity of the acid part. Finally in the bidentate combinations between the phosphoserines and the glutamic acids, the rigidity is also increased, in most of them, this increase in rigidity is comprised to the coordination zone.

3.2. Binding free energies in solution

The stability of the complex formation was studied using the following ligand-substitution reaction:

$$[Al(H_2O)_6]^{3+}_{(aq,1M)} + [L(H_2O)_x]^{k-}_{(aq,1M)} \rightarrow [Al(H_2O)_{6-m}L(H_2O)_{x-m}]^{3-k}_{(aq,1M)} + 2mH_2O_{(aq,1M)} \quad (1)$$

where L is the phosphorylated NF13 with (k; x) = (1, 7) for the NF13_S2P and NF13_S7P peptides, and with (k; x) = (3, 9) for the NF13_S(2,7)P peptide, m depends on the coordination mode, 1 for the monodentates, 2 for the bidentates and 3 for the tridentates. The free energy in solution corresponding to the aluminum binding to the peptides is calculated as follows:

$$\begin{aligned}\Delta G_{aq} &= G_{aq}([Al(H_2O)_{6-m}L(H_2O)_{x-m}]^{3-k}_{(aq,1M)}) \\ &+ 2mG_{aq}(H_2O_{(aq,1M)}) - G_{aq}([Al(H_2O)_6]^{3+}_{(aq,1M)}) \\ &- G_{aq}([L(H_2O)_x]^{k-}_{(aq,1M)}) + \Delta nRTln(24.26) \\ &+ 2mRTln(55.34)\end{aligned} \quad (2)$$

where the last two terms account for the volume change due to the transformation from 1 atm to 1 M in solution, Δn indicates the change in the number of species in the reaction, and the concentration of 55.34 M of water in liquid water. [71] The enthalpy values were also computed excluding the last term. The inclusion of water molecules in the phosphorylated peptides was to avoid proton transfer and changes in the coordination mode due to the unprotected charged residues.

The binding energies of aluminum complexes with the three phosphorylated peptides are displayed in Fig. 7; for clarity only the numbers of the complexes are shown; the label indicating at which peptide belongs was deleted. The results for the NF13_S2P peptides (black circles) show that the monodentate coordination only with the phosphoserine (**1**) leads to the lowest affinity energies, and therefore, it is the least favored possibility. However, the bidentate combination between the S2P and E10 is the complex most stable energetically (**2**), while the other two bidentate combinations are less favorable (**3**, and **4**).

On the contrary, in the case of the NF13_S7P peptides (red squares), the monodentate complex is the thermodynamically most stable (**1**),

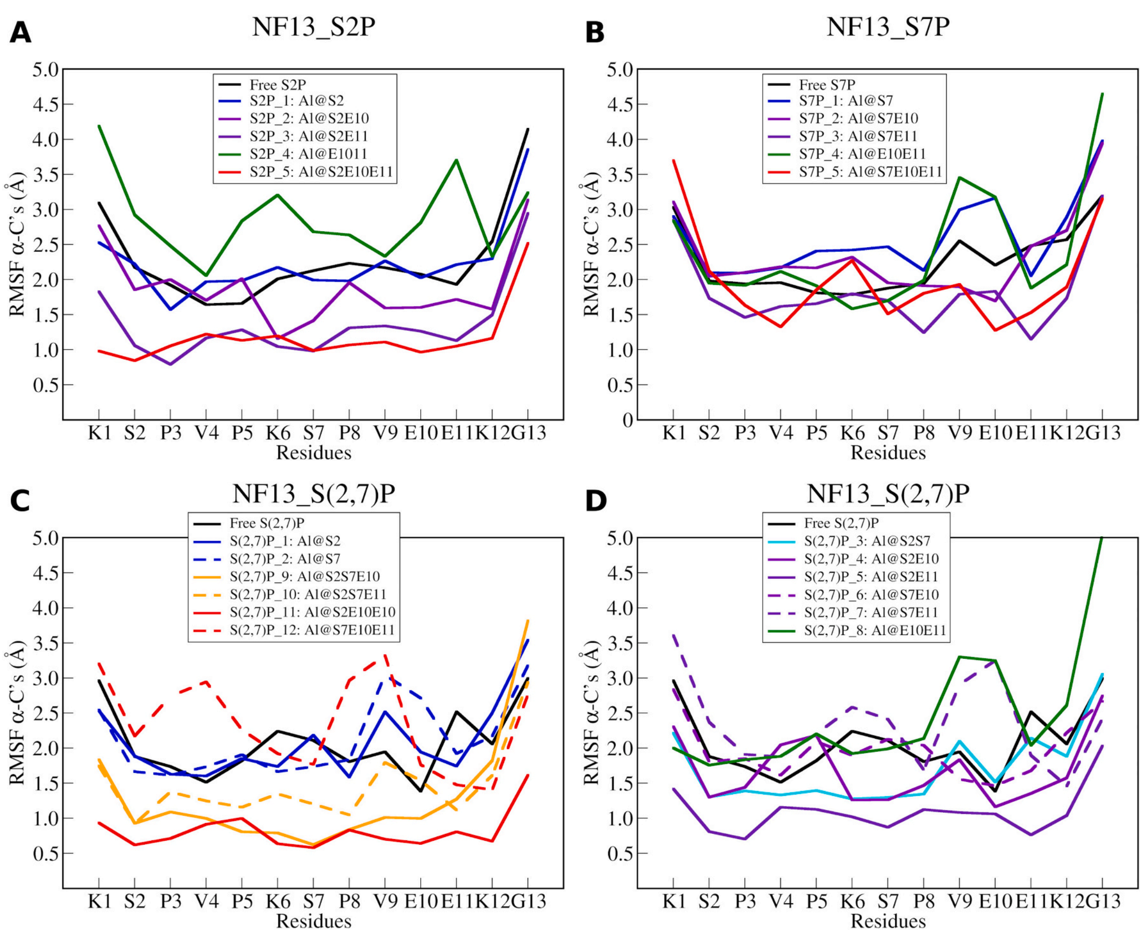

Fig. 6. Root mean square fluctuations of Cα's in each of the phosphorylated complexes, as calculated from the MD simulations.

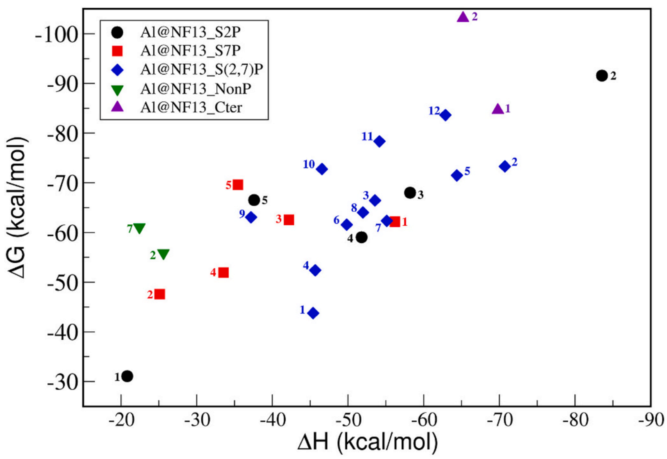

Fig. 7. Aluminum binding energies with the KSPVPKSPVEEKG peptides. The numbers correspond to the complexes showed in the respective figures. Black circles: Aluminum coordinated to the NF13 with S2 phosphorylated (structures in Fig. 4). Red squares: Aluminum coordinated to the NF13 with S7 phosphorylated (structures in Fig. 5). Blue diamonds: Aluminum coordinated to the NF13 with both S2 and S7 phosphorylated (structures in Fig. 6). Green down triangles: Aluminum coordinated to the NF13 non-phosphorylated (structures in Fig. 8). Purple up triangles: Aluminum coordinated to the NF13 non-phosphorylated and with the unprotected C-terminal (structures in Fig. 9).

whereas the bidentate combinations of S7P and E10 (**2**), and both glutamic acids (**4**) show significantly lower binding energies.

In the case of the NF13_S(2,7)P peptides (blue diamonds), the lowest affinity is obtained for the monodentate coordination to S2P (**1**), whereas the coordination to S7P would be energetically more stable (**2**), as in the previous cases.

Regarding the bidentate modes, the coordination to both phosphoserines (**3**) or to both glutamic acids (**8**), they have very similar binding energies, but with some preference for the phosphorylated groups; while the combinations between a phosphoserine and any of the glutamates show a slight preference for E11 (**5** and **7**), preference that is only reversed in the case of the NF13_S2P peptide.

Finally, all the tridentate modes are grouped in the top of the graph, and we can say that the coordination to a phosphoserine and two glutamic acids (**11** and **12**) are slightly favored over that of two phosphoserines and glutamic acid (**9** and **10**). In these complexes, it is clear the effect of the phosphorylation of the peptides; when the peptide has only one phosphoserine to coordinate aluminum in a tridentate fashion the complexes are less favored energetically, structures **5** (red and black) respect to **11** and **12**.

The affinity of the non-phosphorylated peptide was studied with the aluminum coordinated to two glutamic acids and one deprotonated serine (Fig. 8). The binding energies were computed using the eq. 2 with (k; x) = (0, 6).

The calculated energies show less affinity respect to the phosphorylated counterparts (Fig. 7, green down triangles), which is in concordance with the fact that phosphorylation increase the aluminum affinity. On the other hand, we also tried a bidentate mode in which the serines were not deprotonated and the interaction of aluminum is through the two glutamic acids, which gave consistently an even lower interaction, namely, $\Delta G = -22.48$ kcal/mol and $\Delta H = -5.41$ kcal/mol indicating that such coordination would be unlikely.

The fact that phosphorylation clearly increases the affinity of these peptides towards aluminum binding is in agreement with previous experimental and theoretical data. [57,72] However, it seems contradictory with the early work of Hollósi *et at.* [73] which suggested a high affinity of aluminum towards the formation of intrachain complexes with non-phosphorylated NF13 peptides. The authors suggested that the resultant structure could be stabilized by the formation of a salt bridge between the C-terminal acetate group and the lysine and have a tetrahedral coordination. Following that suggestion, we have estimated the stability of this type of structures, considering the NP_7 structure, unprotecting the C-terminal residue and coordinating the aluminum to the deprotonated serine, the two glutamic acids and to the carboxylate from the glycine. Two structures were optimized, one with two water molecules to complete an hexacoordination mode (SG_1), and another tetracoordinated structure without water molecules (SG_2). After the optimization procedure, the SG_1 structure becomes pentacoordinated, loosing the direct interaction with E11, while the SG_2 structure maintains its initial tetracoordination (Fig. 9). In both structures the salt bridge between the terminal carboxylate and the lysine is maintained. The binding energies were computed using the eq. 2 with (k; x) = (1,2). The results show that these structures have high affinities, with SG_2 favored over the other coordination modes, while SG_1 competes with the other tridentate structures (Fig. 7, purple triangles).

In summary, taking into account our estimation of ΔG energies there is a clear increase in stability as we move from monodentate < bidentate < tridentate coordination modes. This trend highlights the ability of aluminum to act as a bridging agent between different residues of the peptide, introducing constraints in the conformational space accessible to the peptide. However, an effect in the binding free energy that we cannot account for with the current level of detail of our calculations is the conformational entropy loss due to these conformational constraints introduced by the binding of aluminum at more than one position of the peptide. The resulting destabilization will be

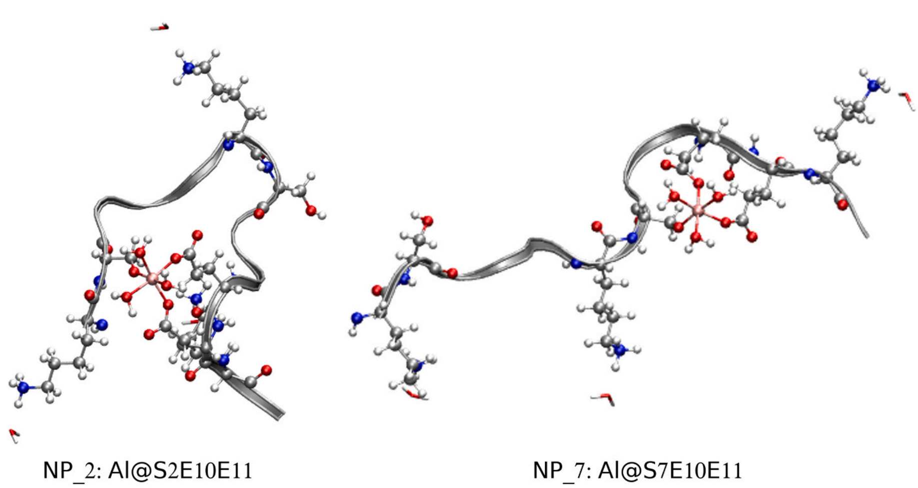

Fig. 8. Optimized structures of the aluminum complexes with the non-phosphorylated NF13 peptide and the deprotonated serines.

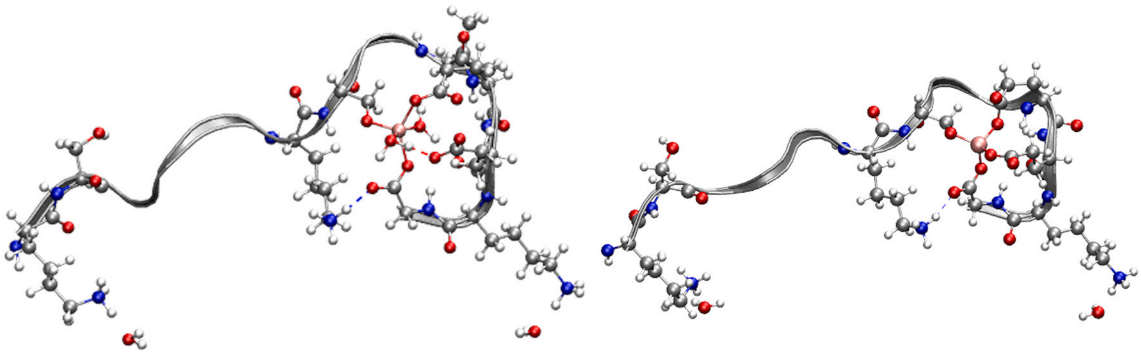

Fig. 9. Optimized structures of the aluminum complexes with the unprotected C-terminal of the NF13-peptide and S7 deprotonated.

trivially larger for tridentate complexes than for bidentate ones.

A second order effect in the free energy will be related to the positions where binding occurs. In this sense, we expect the conformational entropy loss to be larger when aluminum is bound to S2P and E11 than when it is bound to S7P and E10, simply because the arrested fragment is longer in the former case. Although it is difficult to estimate this entropic cost, one possible rule may come from loop insertion experiments in proteins, where each additional aminoacid in an arrested loop destabilized these proteins between 0.1 and 0.26 kcal/mol. [74] Clearly, this will be outweighed by the great enthalpic decrease due to the additional favorable interactions with the metallic cation.

3.2.1. Binding energies at physiological pH

To study the effect of the pH on the interaction of aluminum with this kind of peptides, [72] we have considered the KSPV phosphorylated tetramer as ligand (L), with just one coordination site for aluminum. To avoid proton transfer or bicoordination to the phosphate group, water molecules were added to the charged groups. The calculations were done using Gaussian09 package, [75] at B3LYP/6–31 + G* level, [68,69] solvent effects were taken into account with the IEFPCM implicit method. [76] Using this model, we have computed the binding energies, taking into account the following reactions in which different hydrolytic species of aluminum are considered. These species are the dominant mononuclear aluminum hydrolytic species at different pH's, namely, $[Al(H_2O)_6]^{3+}$ is the dominant species at pH's lower than 5, and $[Al(OH)_4]^-$ at pH's higher than 7. [77]

$$[Al(H_2O)_6]^{3+} + [L(H_2O)_3]^- \rightarrow [Al(H_2O)_5L(H_2O)_2]^{2+} + 2H_2O \quad (3)$$

$$[Al(H_2O)_5OH]^{2+} + (L(H_2O)_3)^- \rightarrow [Al(H_2O)_4L(H_2O)_2OH]^+ + 2H_2O \quad (4)$$

$$[Al(H_2O)_4(OH)_2]^+ + [L(H_2O)_3]^- \rightarrow [Al(H_2O)_3L(H_2O)_2(OH)_2] + 2H_2O \quad (5)$$

$$[Al(H_2O)_3(OH)_3] + [L(H_2O)_3]^- \rightarrow [Al(H_2O)_2L(H_2O)_2(OH)_3]^- + 2H_2O \quad (6)$$

Table 2
Free energies and enthalpies in kcal/mol for the Al–KSPV complexes, depending on the deprotonated aluminum species to simulate the pH. Computed at B3LYP/6–31 + G* level using the Eq. (2).

Complex	ΔG_{aq}	ΔH_{aq}
3 $[Al(H_2O)_5L(H_2O)_2]^{2+}$	−40.75	−36.09
4 $[Al(H_2O)_4L(H_2O)_2OH]^+$	−34.75	−27.50
5 $[Al(H_2O)_3L(H_2O)_2(OH)_2]$	−16.15	−6.81
6 $[Al(H_2O)_2L(H_2O)_2(OH)_3]^-$	−1.73	5.86
7 $[AlL(H_2O)_2(OH)_3]^-$	29.29	35.46

$$[Al(OH)_4]^- + [L(H_2O)_3]^- \rightarrow [AlL(H_2O)_2(OH)_3]^- + H_2O + OH^- \quad (7)$$

Table 2 shows the values of the complexation energies of the aluminum with the KSPV peptide according to the reactions (3)–(7).

Clearly, the affinity of aluminum for the peptide decreases as the pH increases, i.e. as the aluminum coordination sphere is saturated by OH^- groups. Nevertheless, the data show that the aluminum-peptide complexes can be formed departing from different aluminum hydrolytic species (left side of reactions (3)–(6)), as long as one water molecule is displaced from the first coordination shell of aluminum. The values of the reaction (3) are close to the monodentate complexes with the NF13 peptides. However, at the limit of OH^- saturation, namely for $[Al(OH)_4]^-$ species, the interaction is clearly endothermic, since the coordination with the phosphate group implies the release of an hydroxide (reaction (7)), preventing the formation of these Al-Peptide complexes.

In this sense, the presence of hydroxides in the first coordination shell of aluminum has a protective effect precluding its interaction with these peptides. Similar behavior has been found for other ligands like the superoxide. [11,78] This is also coherent with the findings of Hollender et al. [79] that established that for a single-phosphorylated octapeptide the increase in binding affinity to aluminum by the phosphorylation was not enough to transform the peptide into a good aluminum chelator at neutral pH, a result later corroborated by our own calculations. [57] In this sense, although our binding energies displayed in Fig. 7 are quite favorable, specially for S2P_2 structure ($\Delta G = -86.65$ and $\Delta H = -83.57$ kcal/mol), they are still below of our previous estimates of well known aluminum chelators such as citrate ($\Delta G = -124.9$ kcal/mol), [80] the main LMM chelator in blood serum, and by relevant biophosphates such as 2,3-diphosphoglycerate (2,3-DPG) ($\Delta G = -123.5$ kcal/mol), glucose-6-phosphate ($\Delta G = -117.2$ kcal/mol), and ATP-like triphosphates ($\Delta G = -108.7$). [41,81]

We also analyzed the possibility of sequential deprotonation of the water molecules of complex **3** as the pH is raised, leading to complexes with a number of hydroxides in the first solvation layer of aluminum. To do so, we evaluate the reaction energies corresponding to deprotonation of a water molecule of the aluminum-peptide complex and protonation of a HPO_4^{2-} phosphate (reaction (8)). Notice that the pKa2 of orthophosphoric acid (namely, the one that corresponds to $H_2PO_4^- \rightarrow HPO_4^{2-} + H^+$) is 7.2, and therefore the reaction energies calculated in this way can give an idea of the stability of the complex at a neutral pH.

$$[Al(H_2O)_{5-m}L(H_2O)_2(OH)_m]^{2-m} + HPO_4^{2-} \rightarrow [Al(H_2O)_{4-m}L(H_2O)_2(OH)_{1+m}]^{1-m} + H_2PO_4^{-1} \quad (8)$$

The results collected in Table 3 shows that at neutral pH's these complexes could evolve to Al-peptide complex in which the water

Table 3
Free energies and enthalpies in kcal/mol of the reaction (8). Computed at B3LYP/6–31 + G* level using the Eq. (2) without the last two terms. † Aluminum loses two water molecules from its coordination.

Al–KSPV	HPO_4^{2-}	
Complex	ΔG_{aq}	ΔH_{aq}
3 → **4**	−37.92	−34.14
4 → **5**	−15.15	−13.56
5 → **6**	−5.96	−7.26
6 → **7**†	−29.81	−10.57

molecules bounded to the aluminum are deprotonated and the complex is surrounded with up to three hydroxide groups (complex **6**). This complex further evolves releasing the last two remaining water molecules of the first solvation shell of aluminum, leading to complex **7** in which aluminum is bound tetrahedrically to three hydroxides and to the phosphorylated serine.

Finally, we have estimated the possibility of various aluminum hydrolytic species to bind to the phosphorylated NF13 peptides. The combination of possibilities between different binding modes (monodetates, bidentates and tridentates) and different number of hydrolytic species make very difficult a full systematic calculation in this sense. But based on what we have learned above for the Al–KSPV system, we have selected structures in which aluminum will be bound to four negative charged oxygen atoms. This implies that if the aluminum is bound monodentately, there is the possibility to accommodate three hydroxides, for the bidentates modes two hydroxides and for the tridentate modes only one hydroxide. The reference hydrolytic species therefore has different number of hydroxides as a function of the coordination mode. We also consider that the interaction with the peptide involves the release of water molecules rather than hydroxides. That is, ligands that bind monodentately, bidentately and tridentately would release one, two and three water molecules, respectively. Therefore, taking into account the structures of the Figs. 1–3, the coordination sphere of aluminum was altered to introduce the corresponding hydroxides by deprotonation of selected water molecules and the resultant structures were optimized at B3LYP/6-31G* level. The ΔG_{aq} and ΔH_{aq} for aluminum-NF13 binding were calculated according to the following reactions:

- Monodentate complexes

$$[Al(H_2O)_3(OH)_3] + [L(H_2O)_x]^{k-} \rightarrow [AlL(H_2O)_{x-m}(OH)_3]^{k-} + (3+m)H_2O \quad (9)$$

- Bidentate complexes

$$[Al(H_2O)_4(OH)_2]^{+} + [L(H_2O)_x]^{k-} \rightarrow [AlL(H_2O)_{x-m}(OH)_2]^{1-k} + (4+m)H_2O \quad (10)$$

- Tridentate complexes

$$[Al(H_2O)_5(OH)]^{2+} + [L(H_2O)_x]^{k-} \rightarrow [AlL(H_2O)_{x-m}(OH)]^{2-k} + (5+m)H_2O \quad (11)$$

where L is the phosphorylated NF13 with (k; x) = (1, 7) for the NF13_S2P and NF13_S7P peptides, and with (k; x) = (3, 9) for the NF13_S(2,7)P peptide, and m depends on the coordination mode, 1 for the monodentates, 2 for the bidentates and 3 for the tridentates. Table 4 shows the binding energies for the different coordination modes of the reactions (9)–(11) computed using the Eq. (2) without the last two terms. As previously mentioned, the conformational entropy loss due to the constraints imposed by the multidentate aluminum coordination was not considered in the binding energies.

The results for NF13 peptides are consistent with our previous results for Al-KSPV system, suggesting that the presence of hydroxides has again a damping effect on the affinity between the peptide and aluminum: the more hydroxides in the first solvation layer the lower the affinity. Various aluminum hydrolytic species have still favorable thermodynamics towards Al-NF13 peptide formation, and among the different type of species, tridentate complexes are the most favorable ones, while the formation of some monodentate and bidentate complexes would be enthalpicaly unfavored. Our results suggest that the formation of such complexes can be possible due the affinity of aluminum by the phosphorylated peptides. However, as the pH is raised and there are more hydroxides in the first coordination shell, a reduction in the coordination number of aluminum is predicted, which in turn reduce the capacity of aluminum to act as a bridging agent between different residues of the peptides.

In general, our results emphasize the rich coordination chemistry of aluminum to this type of NF13 peptides with a high variety of potential complexes to be formed and favoring structures and coordination modes in which aluminum acts as an aggregation agent that bridges different sidechains of the peptide, altering at least locally the corresponding secondary structure.

Table 4
Free energies and enthalpies in kcal/mol for the Al–NF13 complexes depending on the pH values. Computed at B3LYP/6-31G* level using the Eq. (2) without the last two terms. Conformational entropy loss not consider in the free energies (see text).

NF13_S2P			NF13_S7P			NF13_S(2,7)P		
	ΔG_{aq}	ΔH_{aq}		ΔG_{aq}	ΔH_{aq}		ΔG_{aq}	ΔH_{aq}
Monodentate complexes								
1	−13.14	22.15	1	−22.24	8.78	1	−19.55	1.47
						2	−15.29	−1.24
Bidentate complexes								
2	−55.61	−15.94	2	−33.54	20.90	3	−36.50	4.65
3	−66.43	−25.16	3	−36.34	16.80	4	−36.68	−1.81
4	−53.00	−13.23	4	−51.70	0.22	5	−28.39	10.19
						6	−42.69	3.71
						7	−40.52	7.92
						8	−53.45	−13.94
Tridentate complexes								
5	−75.73	−4.15	5	−78.07	−6.47	9	−67.44	−4.00
						10	−77.38	−15.90
						11	−84.78	−32.71
						12	−80.53	−17.31

4. Conclusions

In this work we have used classical MD simulations and DFT calculations to study complexes formed by the NF13 peptide, phosphorylated in different positions, and the Al(III) cation. NF13 shows several potential binding sites, which leads to various possible coordination modes (mono, bi and tridentate) with different combination of residues and stable structures. The binding to aluminum induces different types of structural changes depending on which residues are involved in the coordination, which are more important in multidentate binding modes with the S2P residue. In that case, a loop is formed, causing a greater rigidity of the backbone and important local changes in the secondary structure of the peptide. On the other hand, the combinations with S7P tend to anchor the E10 and/or E11 residues in the proximity, leaving the rest of the structure quite flexible. Finally, the coordination of aluminum to both glutamates E10 and E11 leads to the most flexible structures. In summary, our results for the interaction of aluminum with a single NF13 peptide show that this metal can form a high variety of structures. On one hand, it favors intrachain interactions with the formation of a loop mainly when the S2P residue is involved in the binding, and more extended conformations when other residues are involved in the binding. We notice that for the formation of plaques an extended conformation would be needed, which can give rise to stacking through interchain interactions.

Our estimations of binding free energies support the idea that tridentate modes are preferred over bidentate and monodentate ones. On the other hand, the phosphorylation of serines increases the affinity of NF13 towards aluminum, although the calculated interaction energies are still lower than well-known aluminum chelators in biological systems such as citrate, ATP or 2,3-DPG characterized in our group in previous works. [80] Finally, the presence of hydroxides in the first solvation layer of aluminum has a damping effect on the affinity of aluminum for the phosphorylated NF13 peptide, although still there are a number of hydrolytic aluminum species that could favorable interact with the NF13 peptide.

Author contributions

The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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